

Solar based energy transition with zero emission

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Abstract:-

Preventing resource conflicts in the face of increasing global populations and demands in the 21st century is a high priority for the Department of Defence and for common people. All solution options to these challenges should be explored, including opportunities from space. Space-based solar power (SBSP) can be developed and deployed to provide affordable, clean, safe, reliable, sustainable and expandable energy for mankind. The basic idea is very straightforward: place very large solar arrays into continuously and intensely sunlit Earth orbit (1,366 watts/m²), collect gigawatts of electrical energy, electromagnetically beam it to Earth, and receive it on the surface for use either as base load power via direct connection to the existing electrical grid, conversion into manufactured synthetic hydrocarbon fuels, or as low-intensity broadcast power beamed directly to consumers. This sbps based energy transition is briefly reviewed.

Introduction:-

A single kilometer-wide band of geosynchronous earth orbit experiences enough solar flux in one year to nearly equal the amount of energy contained within all known recoverable conventional oil reserves on Earth today. This amount of energy indicates that there is enormous potential for energy security, economic development, improved environmental stewardship, advancement of general space faring, and overall national security for those nations who construct and possess a SBSP capability. NASA and DOE have collectively spent \$80M over the last three decades in sporadic efforts studying this concept (by comparison, the U.S. Government has spent approximately \$21B over the last 50 years continuously pursuing nuclear fusion).

Commercial Baseload SBSP must be able to compete with Land Based (Terrestrial) Solar Power, Wind Power, Biofuels, Fossil Fuels, Nuclear Fission and with future sources of energy which have not yet been developed. Commercial Baseload SBSP must be able to compete with Land Based (Terrestrial) Solar Power, Wind Power, Biofuels, Fossil Fuels, Nuclear Fission and with future sources of energy which have not yet been developed.

This chart compares a number of sources of power with SBSP in terms of their safety, reliability, ability to provide baseload power and whether they qualify as clean and green.

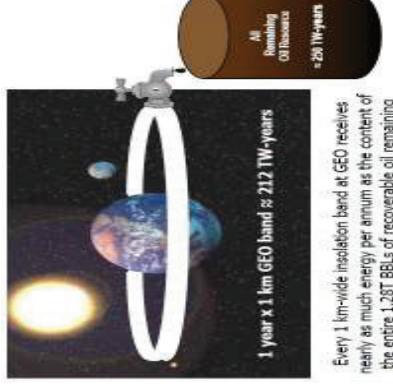
Source	Clean	Safe	Reliable	Base-load
Fossil Fuel	No	Yes	Decades remaining	Yes
Nuclear	No	Yes	Fuel Unlimited	Yes
Wind Power	Yes	Yes	Intermittent	No
Ground Solar	Yes	Yes	Intermittent	No
Hydro	Yes	Yes	Brought Complex Scheduling	No
Bio-fuels	Yes	Yes	Unlimited Qty - Complex w/ food	No
Space Solar	Yes	Yes	Yes	Yes

Other speculative energy sources which may come into importance in the future could include fusion. Fusion would be the most comparable to SBSP as it also offers access to, for all immediate purposes, an unlimited supply of energy. The interest of the public and of investors in alternative energies and fuels is rising rapidly.

Study Objective:-

Equitable access to sufficient quantities of clean, reliable, and affordable energy fundamentally enables the technical and policy solutions that can prevent future resource conflicts while still providing opportunities for

prosperous growth. Every energy resource opportunity, including those from space, must be fully explored to determine its ability to contribute toward solving mankind's looming energy supply and security dilemma.



A single kilometer-wide band of geosynchronous earth orbit experiences enough solar flux in one year (approximately 212 terawatt-years) to nearly equal the amount of energy contained within all known recoverable conventional oil reserves on Earth today (approximately 250 TW-yrs). The enormous potential of this resource demands an examination of mankind's ability to successfully capture and utilize this energy within the context of today's technology, economic, and policy realities, as well as the expected environment within the next 25 years. Study of space-based solar power (SBSP) indicates that there is enormous potential for energy security, economic development, advancement of general space faring, improved environmental stewardship, and overall national security for those nations who construct and possess such a capability.

First originated as an idea in 1968 and later patented by Dr. Peter Glaser, Space-Based Solar Power captures sunlight on orbit where it is constant and stronger than on Earth, and converts it into coherent radiation that is beamed down to a receiver on Earth.

1). GEO Solar Power Architecture

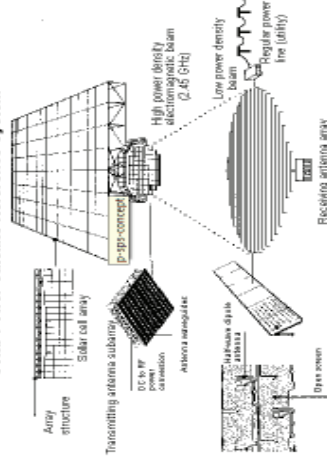
2). Lunar Solar Power Architecture



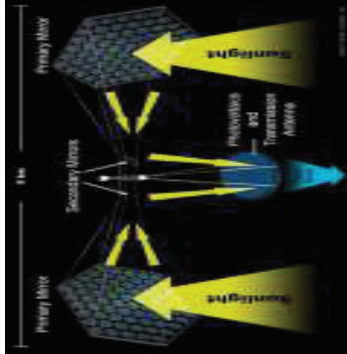
Lunar Solar Power Architecture

Two basic architectures exist placement of collectors in Earth orbit [geostationary orbit (GEO), medium-Earth orbit (MEO), or low-earth orbit (LEO)], or placement of collectors on the surface of the Moon. Two basic methods exist for capturing the energy: photovoltaic or solar dynamic. Finally two basic methods of beaming the power down exist: via coherent radio waves, or via coherent visible or infrared light.

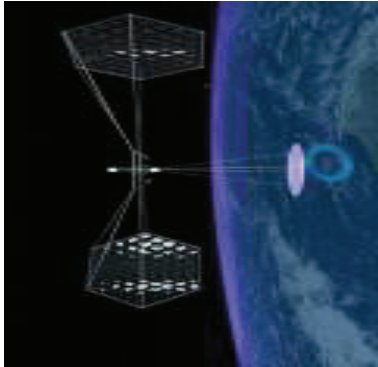
Solar Power Satellite Reference System



Typical reference designs involved a satellite in geostationary orbit, several kilometers on a side, that used photovoltaic arrays to capture the sunlight, then convert it into radio frequencies of 2.45 or 5.8 GHz where atmospheric transmission is very high, that were then beamed toward a reference signal on the Earth at intensities approximately 1/6th of noon sunlight. The beam was then received by a rectifying antenna and converted into electricity for the grid, delivering 5-10 gigawatts of electric power.



a).Collection of solar energy in the space



b).Sending of the energy from space towards earth

Fig :- Current Design Proposals

Mission

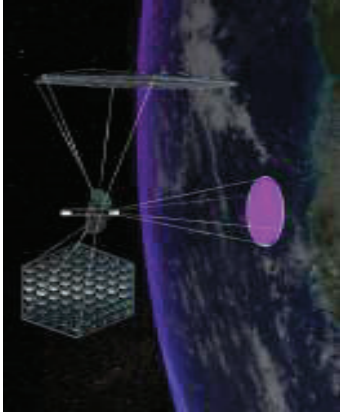
Determine the political, scientific, technical, logistical, and commercial feasibility of space-based solar power collection and distribution in the 21st Century as a contributing source of clean energy to national power grids and smaller niche applications. Discuss significant capabilities, limitations, and alternatives. Identify challenges that must be overcome, and suggest a research and development investment plan that incrementally retires risk on the path to fielding an operational system.

Nominal Goals

Operational: Initial capability to deliver 5-10 MWe of power to remote locations

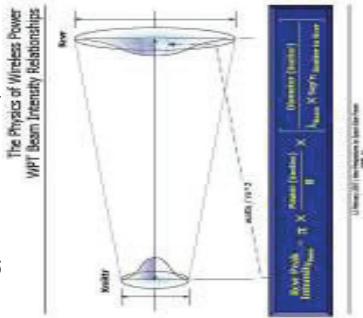
Strategic: Capacity to deliver 10% of required energy in the upcoming years.

FOCUS AREAS - SCIENCE & TECHNOLOGY



Whether Lunar, LEO, or GEO based, whether at radio or optical frequencies, Space-Based Solar Power is an immense and cross-disciplinary technological challenge, requiring contributions from many diverse disciplines. A Space Solar Power Satellite is a huge construction, requiring expertise in large structures, control and orbital dynamics. Its construction requires sophisticated techniques and careful logistics. Its power generation, be it solar-dynamic, thermo-electric, or photovoltaic, requires expertise in solid-state electronics, optics and thermodynamics. SPS power collection and distribution require electrical engineering, and

the conversion of the electrical power generated in space into a coherent wireless power beam requires significant knowledge of the physics of electromagnetic wave propagation. Each of these challenges in turn also depends on competence in advanced material sciences. In all areas, there has been tremendous progress in 21st century. In evaluating the long term potential of SBSP, it is instructive to consider that global space utilities that are taken for granted today such as Global Positioning System (GPS), and Geostationary Satellites ("The Clarke Belt") were once scoffed at and treated as mere science fiction until technology was able to catch up.



- The microwave beams are constant and conversion efficiencies high, they can be beamed at densities substantially lower than that of sunlight and still deliver more energy per area of land usage than terrestrial solar energy. The peak density of the beam is likely to be significantly less than noon sunlight, and at the edge of the rectenna equivalent to the leakage allowed and accepted by hundreds of millions in their microwave ovens. This low energy density and choice of wavelength also means that biological effects are likely extremely small, comparable to the heating one might feel if sitting some distance from a campfire.

- The physics of electromagnetic energy beaming is uncompromising, and economies of scale make the beam very unsuitable as a "secret" weapon. Concerns can be resolved through an inspection regime and better space situational awareness capabilities. The distance from the geostationary belt is so vast that

beams diverge beyond the coherence and power concentration useful for a weapon. The beam can also be designed in such a manner that it requires a pilot signal even to concentrate to its very weak level. Without the pilot signal the microwave beam would certainly diffuse and can be designed with additional failsafe cut-off mechanisms. The likelihood of the beam wandering over a city is extremely low, and even if occurring would be extremely anti-climactic.

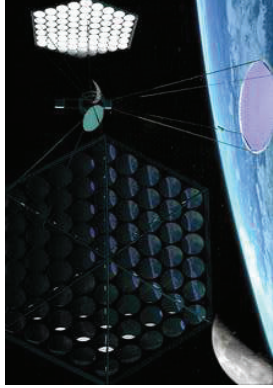
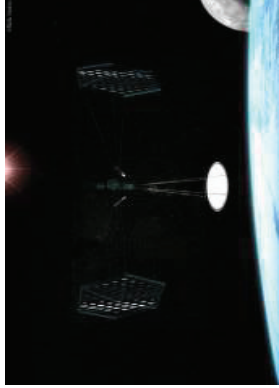


Fig:-Collection of solar energy in space using large collector.

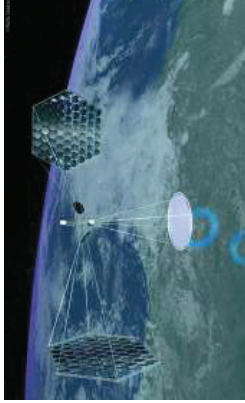
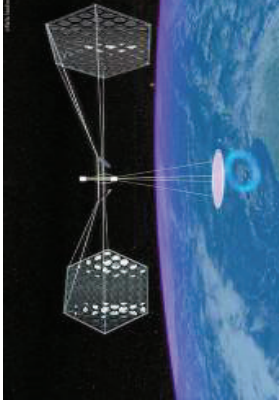


Fig:-Demonstrate the transition of energy towards earth

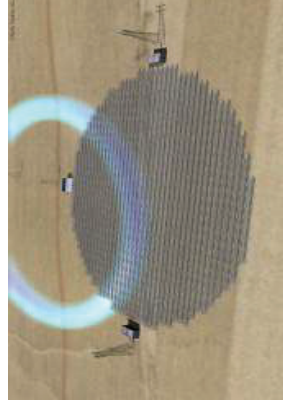
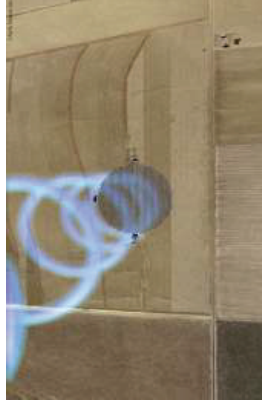


Fig:-Earth ground station



Fig:-Powering the cities and industries.